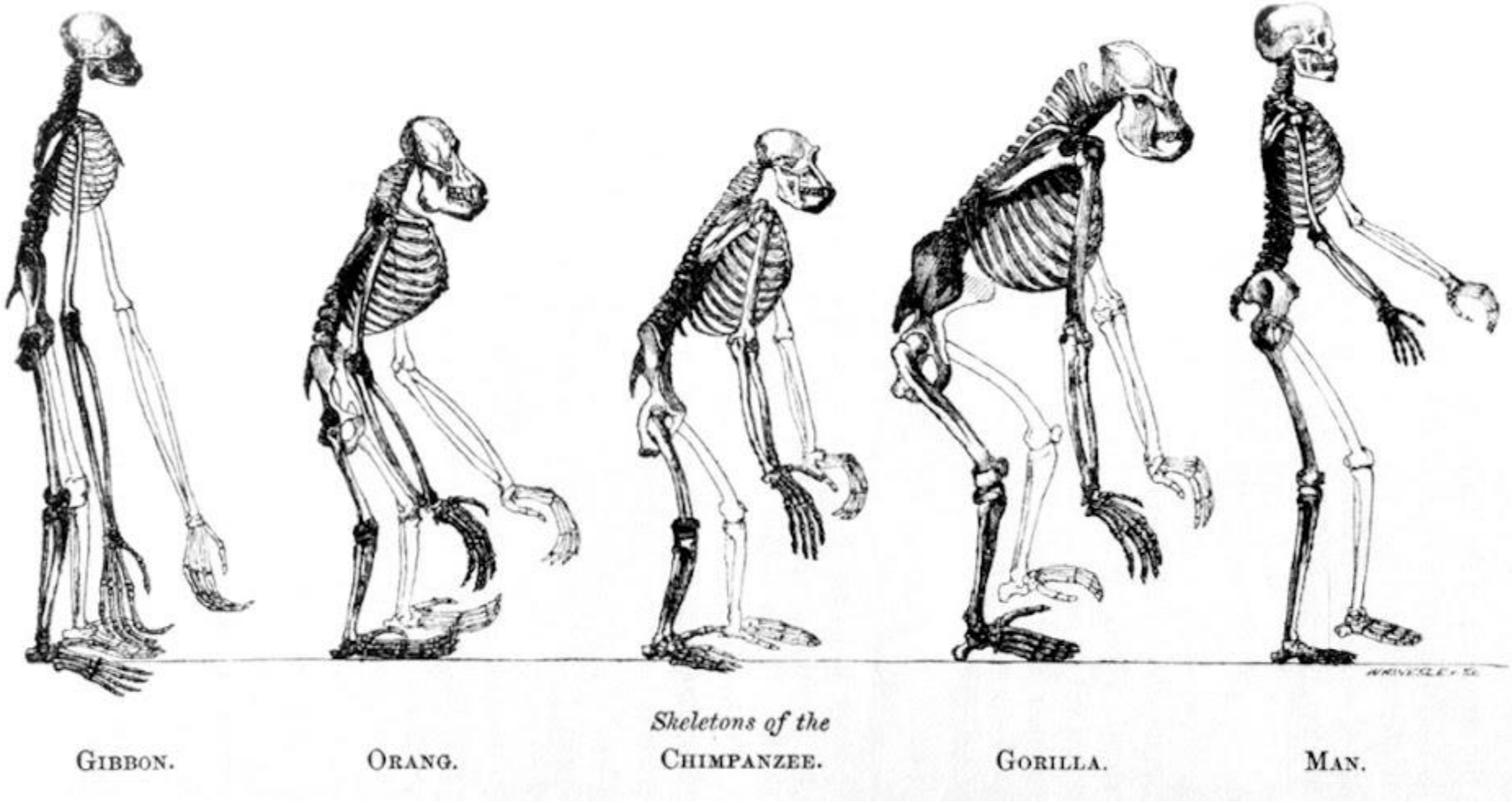
How to tell a story with charts in five charts: A newsroom taxonomy



https://en.wikipedia.org/wiki/Man%27s_Place_in_Nature

My name is Ben

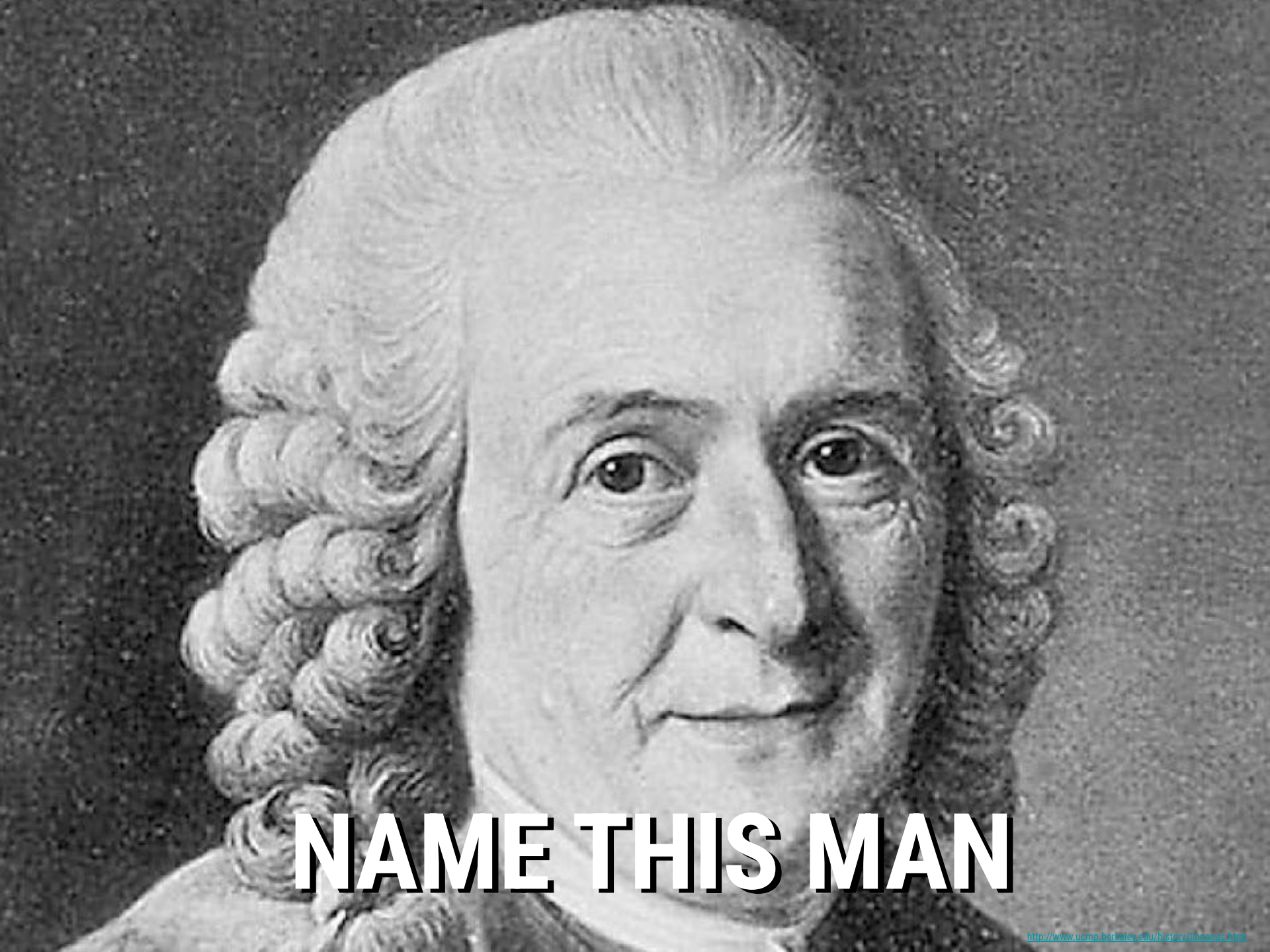
I sometimes go by @palewire

My name is Ingrid

My nickname is Yeye. My handle: @yeyelobet

Find our slides here

bit.ly/ire-sandiego-2016


NAME THIS MAN
http://www.ucmp.berkeley.edu/history/linnaeus.html

CAROLI LINNÆI

I. QUADRUPEDIA.

Corpus hirsutum. *Pedes* quatuor. *Femina* viviparæ, lactiferæ.

ANTHROPOMORPHA. *Dentes primores 4. utrinque; vel nulli.*	Homo.	Nosce te ipsum.	H { Europæus albesc. / Americanus rubesc. / Asiaticus fuscus. / Africanus nigr.
	Simia.	ANTERIORES. POSTERIORES. *Digiti* 5. . . . . 5. Posteriores anterioribus similes.	Simia cauda carens. Papio. Satyrus. Cercophithecus. Cynocephalus.
	Bradypus.	*Digiti* 3. vel 2. . . 3.	Ai. *Ignavus.* Tardigradus.
FERÆ. *Dentes primores 6. utrinque: intermedii longiores: canini acuti. Pedes multifidi, unguiculati.*	Ursus.	*Digiti* 5. . . . . 5. Scandens. *Mamma* 4. (Ald.) *Calcaneis* insistit. *Pollex* extus positus.	Ursus. Costi *Merg.* Wickhead *Angl.*
	Leo.	*Digiti* 5. . . . . 4. Scandens. *Mamma* 2. ventralis. *Lingua* aculeata.	Leo.
	Tigris.	*Digiti* 5. . . . . 4. Scandens. *Mamma* 4 umbilicales. *Lingua* aculeata.	Tigris. Panthera.
	Felis.	*Digiti* 5. . . . . 4. Scandens. *Mamma* 8. sc. 4. pect. 4. abdom. *Lingua* aculeata.	Felis. Catus. Lynx.
	Mustela.	*Digiti* 5. . . . . 5. Scandens. *Dentes* molares 4. utrinque.	Martes. Zibellina. Viverra. Mustela. Putorius.
	Didelphis.	*Digiti* 5. . . . . 5. *Mamma* 8. intra bursulam abdomin.	Philander. *Possum.*
	Lutra.	*Digiti* 5. . . . . 5. Palmipes.	Lutra.
	Odobænus.	*Digiti* 5. . . . . 5. Palmipes. *Dentes* intermedii superiores longissimi.	Rosm. *Morsus.*
	Phoca.	*Digiti* 5. . . . . 5. Palmipes. *Mamma* duæ umbilicales.	Canis marinus.
	Hyæna.	*Digiti* 4. . . . . 4. *Collum* superne jubatum. *Cauda* brevis.	Hyæna *Veter.* Pivam *Lonicini nuper vidit et descripsit* ARTED.
	Canis.	*Digiti* 5. . . . . 4. *Mamma* 10. sc. 4. pect. 6. abdom.	Canis. Lupus. Sequilachi. Vulpes.
	Meles.	*Digiti* 5. . . . . 5. *Ungues* medii digitis ipsis longiores. *Corpus* superne albicat: inferne nigricat.	Taxus. Zibetha.
	Talpa.	*Digiti* 5. . . . . 5. *anteriores* maximi.	Talpa.
	Erinaceus.	*Digiti* 5. . . . . 5. *Spinis* vel *setis* squamosis munitus.	Echinus terrestris. Armadillo.
	Vespertilio.	*Digiti* 5. . . . . 5. *Pes anticus* in alam expansus. *Mamma* 2. pectorales.	Vespertilio. Felis volans *Seb.* Canis volans *Seb.* Glis volans *Seb.*
GLIRES. *Dentes primores 2. utrinque. Pedes multifidi.*	Hystrix.	*Aures* humanæ. *Corpus* spinosum.	Hystrix.
	Sciurus.	*Digiti* 4. . . . . 5. *Cauda* longissima laxigera.	Sciurus. . . . volans.
	Castor.	*Digiti* 5. . . . . 5. palmipes postice. *Cauda* horizontalis, plana, nuda.	Fiber.
	Mus.	*Digiti* 4. . . . . 5. *Cauda* teres, squamosa, hirsuta.	Rattus. Mus domesticus. . . . bacchiarus. . . . macrourus. Lemures. Marmota.
	Lepus.	*Digiti* 5. . . . . 4. *Cauda* brevissima, villosa.	Lepus. Cuniculus.

II. AVES.

Corpus plumosum. *Alæ* duæ. *Pedes* duo. *Rostrum* osseum. *Femina* oviparæ.

ACCIPITRES. *Rostrum uncinatum.*	Psittacus.	*Digiti pedis antici* 2. *postici* 2.	Psittacus.
	Strix.	*Digiti pedis antici* 3. *posticus* 1. *quorum extimus retrorsum flexilis.*	Bubo. Otus. Noctua. Ulula.
	Falco.	*Digiti pedis antici* 3. *posticus* 1.	Aquila. Vultur. Buteo. Falco. Cyanopus. Milvus. Lanius. Pygargus. Nisus. Tinnunculus.
PICÆ. *Rostrum* superne compressum, convexum.	Paradisæa.	*Pennæ* 2. longissimæ, singulares, nec alæ, nec uropygio infixæ.	Manucodiata. Avis Paradisiaca.
	Coracias.	*Pes* 4dact. *Rostrum* exteriores gradatim breviores.	Pica.
	Corvus.	*Pes* 4dact. *Rostrum* æquales.	Corvus. Cornix. Monedula. Lupus. Glandaria. Caryocatactes.
	Cuculus.	*Digiti pedis antici* 2. *postici* 2. *Rostrum* leve.	Cuculus. Torquilla f. Jynx.
	Picus.	*Digiti pedis antici* 2. *Rostrum* angularem.	Picus niger. . . . viridis. . . . varius.
	Certhia.	*Pes* 4dact. *Rostr.* gracile incurvum.	Certhia.
	Sitta.	*Pes* 4dact. *Rostr.* triangulare.	Picus cinereus
	Upupa.	*Pes* 4dact. *Caput* plumis cristatum.	Upupa.
	Ispida.	*Pes* 4dact. cujus digitus extimus medio adnectitur tribus articulis.	Ispida. Merops.
MACRORHYNCHÆ. *Rostr. longiusc. acut.*	Grus.	*Caput* cristatum.	Grus.
	Ciconia.	*Caput* plani, subrotundi.	Ciconia.
	Ardea.	*Ungues* medius inferne serratus.	Ardea.
ANSERES. *Os denato-serratum.*	Platelea.	*Rostr.* depresso-planum, *apice* subrot.	Platea.
	Pelecanus.	*Rostr.* depressum, *apice* unguiculato, inferne bursa instructum.	Onocrotalus.
	Cygnus.	*Rostr.* conico-convexum.	Olor. Ansser. Eider. Anas. Bernicla.
	Anas.	*Rostr.* conico-depressum.	Anas fera. Boschas. Penelope. Glaucium. *A. Domest.* Querquedula.
	Mergus.	*Rostr.* cylindriforme, *apice* adunco.	Mergus. Merganser.
	Graculus.	*Rostr.* conicum, *apice* adunco.	Carbo aquat. Graculus aquat.
	Colymbus.	*Rostr.* subulatum. *Pedes* infra æquilibr.	Colymbus. Podiceps. C. minim. Arctica.
	Larus.	*Rostr.* subulatum. *Pedes* in æquilibr.	Cataractæ. Sterna. Larus. Piscator.
SCOLOPACES. *& trans cylindraceo-teretiusculum.*	Hæmatopus.	*Pes* 3dact. *Rostri* apex compressus.	Pica marina.
	Charadrius.	*Pes* 3dact. *Rostri* apex teres.	Pluvialis. Hiaticula.
	Vanellus.	*Pes* 4dact. *Rostrum* digitis brevius. *Caput* pennato cristatum.	Capella.
	Tringa.	*Pes* 4dact. *Rostrum* digitis brevius. *Caput* simplex.	Tringa. Ocyphus. Pugnax. Gallinula.
	Numenius.	*Pes* 4dact. *Rostrum* digitis longius.	Gallinago. Limosa. Arquata. Recurvirostra.
	Fulica.	*Pes* 4dact. *Digiti* membranis suffulti. *Caput* carnoso-cristatum.	Gallinula aquatica.

III. AMPHIBIA.

Corpus nudum, vel squamosum. *Dentes molares* nulli: *reliqui* semper. *Pinnæ* nullæ.

SERPENTIA.	Testudo.	*Corpus* quadrupedum, caudatum, testa munitum.	Testudo testudinaria. . . . terrestris. . . . marina. Lutaria.
	Rana.	*Corpus* quadrupedum, cauda defitutum, squamis carens.	Bufo. Rana arborea. . . aquatica. . . Carolina.
	Lacerta.	*Corpus* quadrupedum, caudatum, squamosum.	Crocodilus. Allegator. Cordylus. Draco volans. Scincus. Salamandra aq. . . terrestris. Chamæleo. Seps. Senembi *Georg.*
	Anguis.	*Corpus* apodum, teres, squamosum.	Vipera. Caecilia. Aspis. Caudisona. Cobras de Cabelo. Angvis Æsculapii. Cenchris. Natrix. Hydrus.

AMPHIBIORUM Classem ulterius continuare noluit benignitas Creatoris; Ea enim si tot Generibus, quot reliquæ Animalium Classes comprehendunt, gauderet; vel si vera essent quæ de Draconibus, Basiliscis, ac ejusmodi monstris et τερατολόγοις fabulantur, certè humanum genus terram inhabitare vix posset.

PARADOXA.

HYDRA corpore anguino, pedibus duobus, collis septem, & totidem capitibus, alarum expers, asservatur Hamburgi, similitudinem referens Hydræ Apocalypticæ à S. JOANNE CAP. XII. & XIII. descriptæ. Eaque tanquam veri animalis speciem plurimis præbuit, sed falso. Natura sibi semper similis plura capita in uno corpore nunquam produxit naturaliter. Fraudem & artificium, cum Ipsi vidimus, dentes Ferino-mustelini, ab Amphibiorum dentibus diverti, facillime detexerunt.

RANA-PISCIS & RANÆ IN PISCEM METAMORPHOSIS valdè paradoxa est, quum Natura mutationem Generis unius in aliam diversæ Classis non admittat. Ranæ, ut Amphibia omnia, pulmonibus gaudent & ossibus spinosis. Pisces spinosi, loco pulmonum, branchiis instruuntur. Ergo lege Naturæ contraria foret hæc mutatio. Si enim piscis hic instructus est branchiis, erit diversus à Rana & Amphibiis. Si verò pulmones, erit Lacerta: nam toto cælo à Chondropterygiis & Plagiuris differt.

MONOCEROS *Veterum*, corpore equino, pedibus ferinis cornu recto, longo, spiraliter intorto, Pictorum figmentum est. MONODON *Artedi* ejusmodi cornu gerit, cæteri verò partibus multum differt.

PELECANUS rostro vulnus infligens femori suo, ut ema

A PERIODIC TABLE OF VISUALIZATION METHODS

Data Visualization
Visual representations of quantitative data in schematic form (either with or without axes)

Information Visualization
The use of interactive visual representations of data to amplify cognition. This means that the data is transformed into an image, it is mapped to screen space. The image can be changed by users as they proceed working with it

Concept Visualization
Methods to elaborate (mostly) qualitative concepts, ideas, plans, and analyses.

Strategy Visualization
The systematic use of complementary visual representations in the analysis, development, formulation, communication, and implementation of strategies in organizations.

Metaphor Visualization
Visual Metaphors position information graphically to organize and structure information. They also convey an insight about the represented information through the key characteristics of the metaphor that is employed

Compound Visualization
The complementary use of different graphic representation formats in one single schema or frame

>☼< C continuum																	>☼< G graphic facilitation
>◎< Tb table	>☼< Ca cartesian coordinates											>☼< Me meeting trace	>☼< Mm metro map	☼ Tm temple	<☼> St story template	>☼< Tr tree	¤ Ct cartoon
>☼< Pi pie chart	>☼< L line chart											>☼< Co communication diagram	>☼< Fp flight plan	>☼< Cs concept sceleton	☼ Br bridge	>☼< Fu funnel	☼ Ri rich picture
>☼< B bar chart	>☼< Ac area chart	>☼< R radar chart cobweb	>◎< Pa parallel coordinates	>◎< Hy hyperbolic tree	>☼< Cy cycle diagram	>☼< T timeline	>☼< Ve venn diagram	<◎> Mi mindmap	<☼> Sq square of oppositions	>☼< Cc concentric circles	>☼< Ar argument slide	>◎< Sw swim lane diagram	>☼< Gc gantt chart	<◎> Pm perspectives diagram	>◎< D dilemma diagram	<☼> Pr parameter ruler	☼ Kn knowledge map
>☼< Hi histogram	>☼< Sc scatterplot	>☼< Sa sankey diagram	>◎< In information lense	>¤< E entity relationship diagram	>☼< Pt petri net	>◎< Fl flow chart	<☼> Cl clustering	>☼< Lc layer chart	>◎< Py minto pyramid technique	>☼< Ce cause-effect chains	>☼< Tl toulmin map	>◎< Dt decision tree	>¤< Cp cpm critical path method	<☼> Cf concept fan	>◎< Co concept map	☼ Ic iceberg	☼ Lm learning map
>☼< Tk tukey box plot	>☼< Sp spectogram	>☼< Da data map	>◎< Tp treemap	>◎< Cn cone tree	>☼< Sy system dyn./ simulation	>◎< Df data flow diagram	<☼> Se semantic network	>◎< So soft system modeling	◎ Sn synergy map	<☼> Fo force field diagram	>¤< Ib ibis argumentation map	>☼< Pr process event chains	>☼< Pe pert chart	<◎> Ev evocative knowledge map	>◎< V Vee diagram	<☼> Hh heaven 'n' hell chart	◎ I informural

Cy Process Visualization

Hy Structure Visualization

☼ Overview
¤ Detail

◎ Detail AND Overview

< > Divergent thinking

> < Convergent thinking

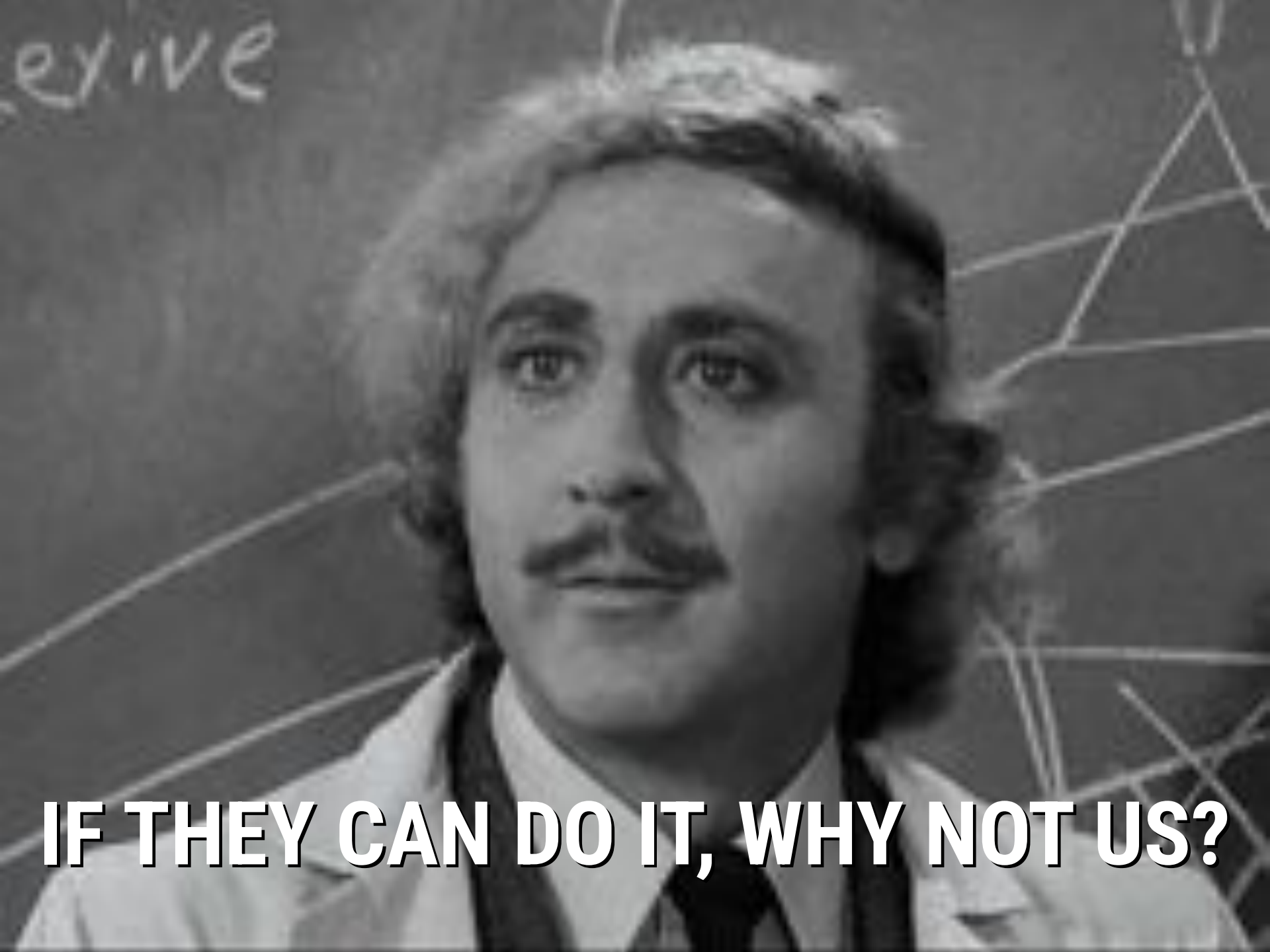
Note: Depending on your location and connection speed it can take some time to load a pop-up picture.

version 1.5

© Ralph Lengler & Martin J. Eppler, www.visual-literacy.org

>☼< Su supply demand curve	>◎< Pc performance charting	>☼< St strategy map	>☼< Oc organisation chart	<¤> Ho house of quality	>☼< Fd feedback diagram	¤ Ft failure tree	>☼< Mq magic quadrant	>☼< Ld life-cycle diagram	>☼< Po porter's five forces	<¤> S s-cycle	>☼< Sm stakeholder map	◎ Is ishikawa diagram	☼ Tc technology roadmap
☼ Ed edgeworth box	>◎< Pf portfolio diagram	☼ Sg strategic game board	>☼< Mz mintzberg's organigraph	<¤> Z zwicky's morphological box	<◎> Ad affinity diagram	¤ De decision discovery diagram	>☼< Bm bcg matrix	>☼< Stc strategy canvas	>☼< Vc value chain	<¤> Hy hype-cycle	>☼< Sr stakeholder rating map	>☼< Ta taps	<¤> Sd spray diagram

http://www.visual-literacy.org/periodic_table/periodic_table.html


IF THEY CAN DO IT, WHY NOT US?

Five graphics

Five categories

Five toolsets

Five tutorials

Five changes

One:

Reporters

Two:

Miners

Three:
Sculptors

Four:
Debaters

Five:

Tailors

YOU READY

One:

Reporters

re·port·er, *noun.* **1.** a stack of maps, charts and other diagrams that summarize and retell a breaking-news event in chronological order.

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MASACRE EN ORLANDO

Masacre en Orlando: reconstrucción gráfica del ataque

Omar Mateen, un ciudadano estadounidense de 29 años, asesinó el domingo de madrugada a 49 personas en un conocido club gay de Orlando. Se atrincheró en la discoteca con un grupo de rehenes. Tres horas después fue abatido por las fuerzas de asalto.

Por: Mariano Zafra, A. Verde, Javier Figueroa y Luis Melgar

Seguir

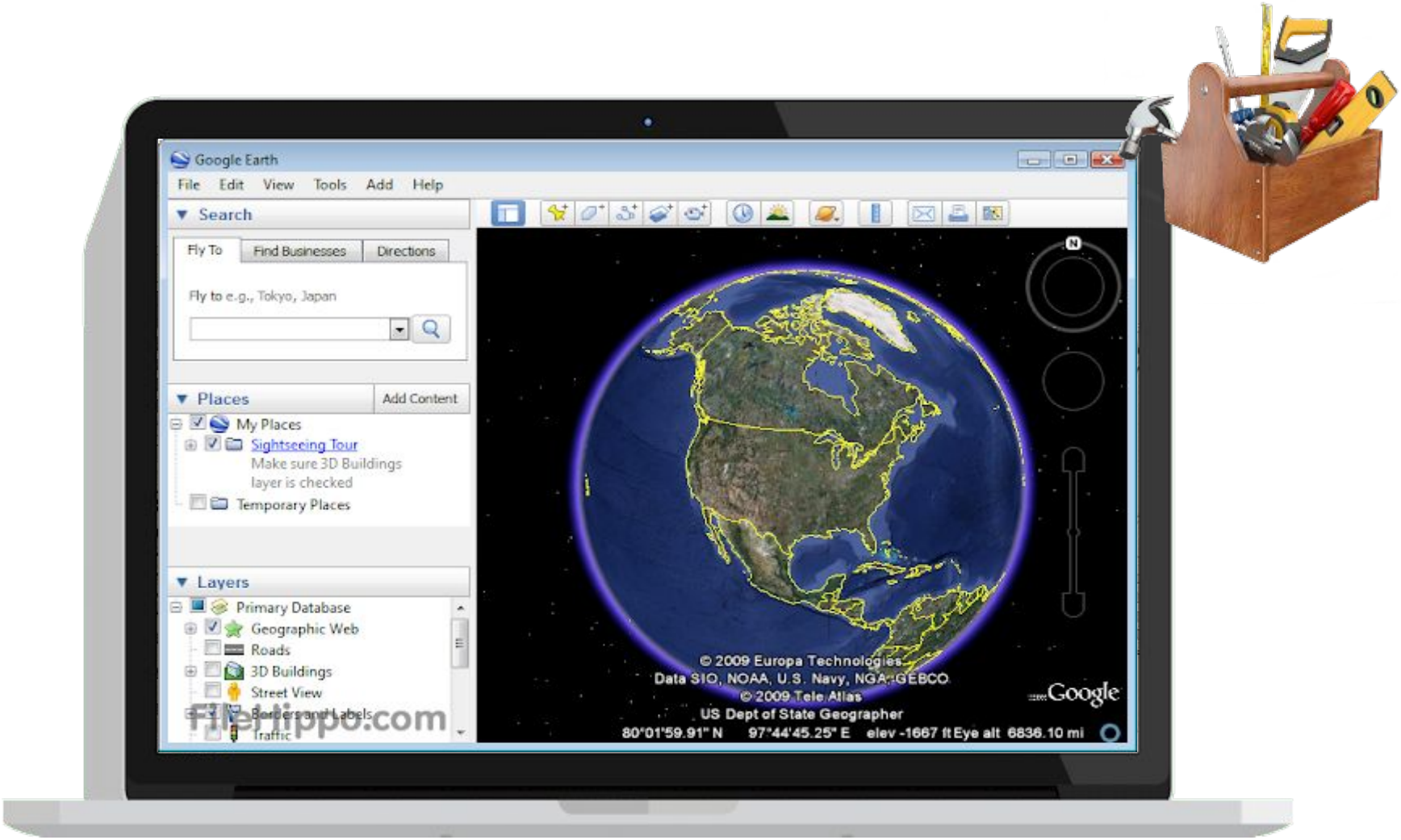
Actualizado: 13 Jun | 6:30 pm

Cronología del ataque

El asaltante, Omar Mateen, un ciudadano norteamericano de 29 años, conduce 120 millas desde la ciudad en la que vivía, Fort Pierce, hasta Orlando.

Antes de las
2:00 am

http://www.univision.com/noticias/masacre-en-orlando/nuevos-detalles-de-la-masacre-en-orlando-reconstruccion-grafica

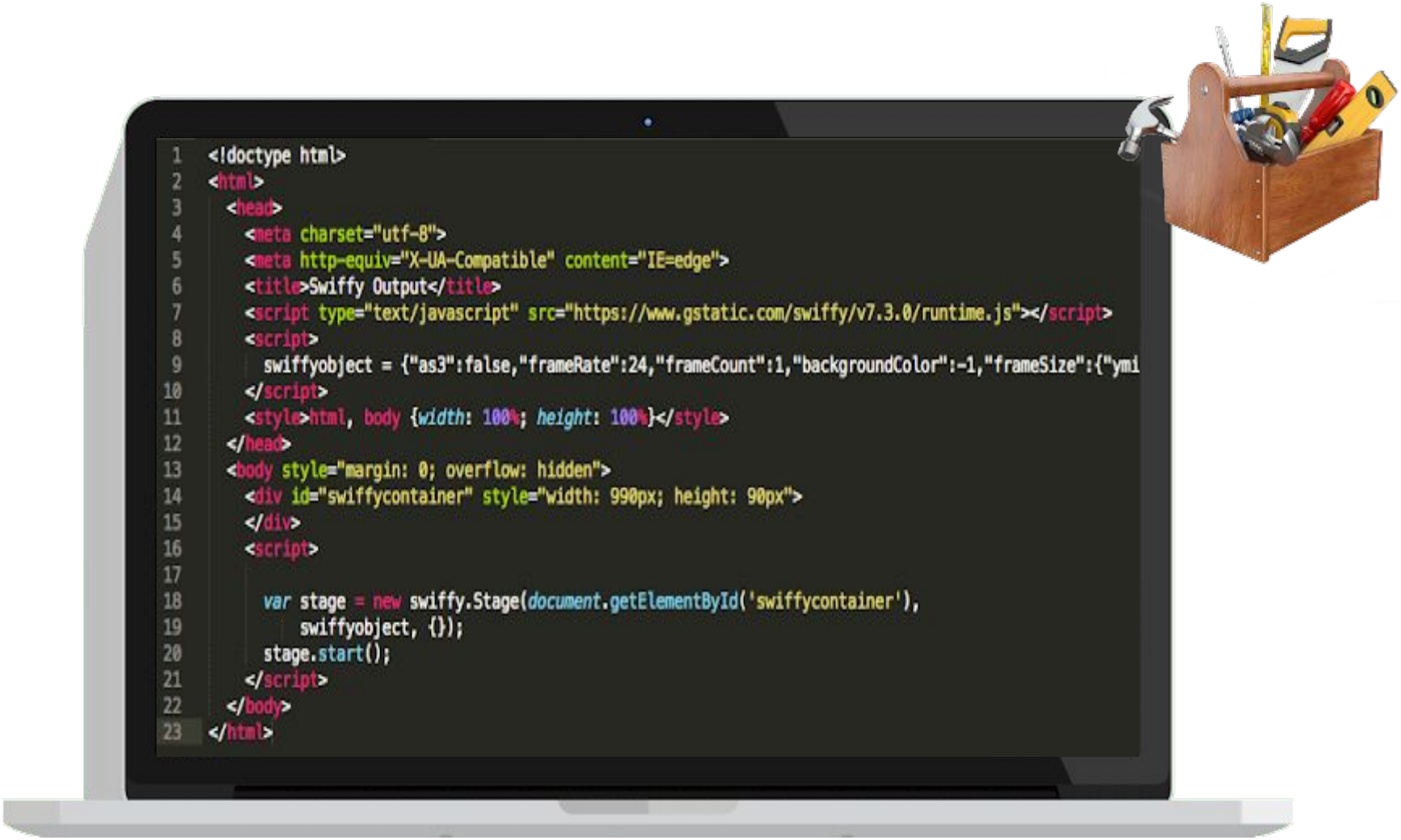


Google Earth

https://www.google.com/earth/

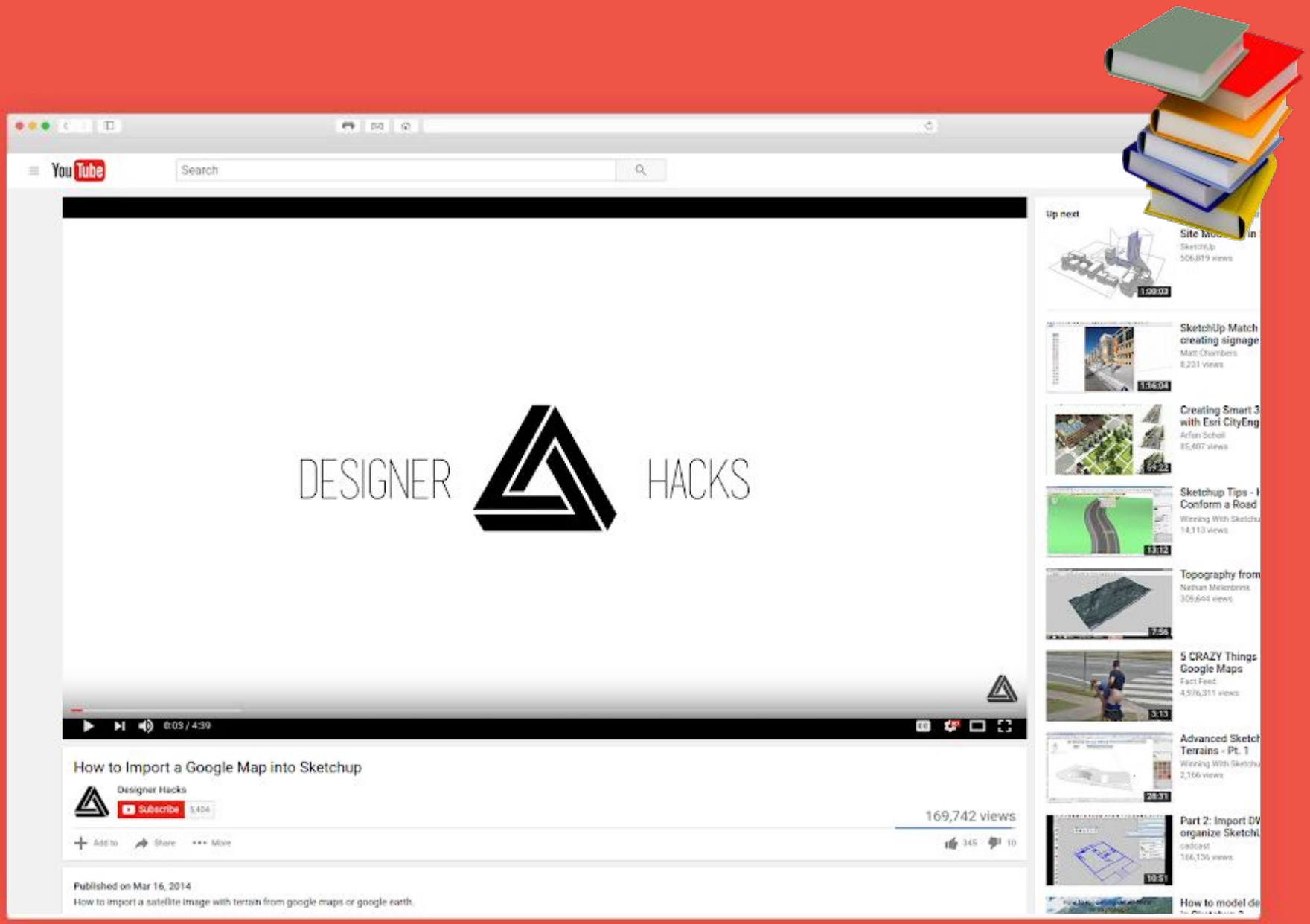
SketchUp

http://www.sketchup.com/products/sketchup-pro

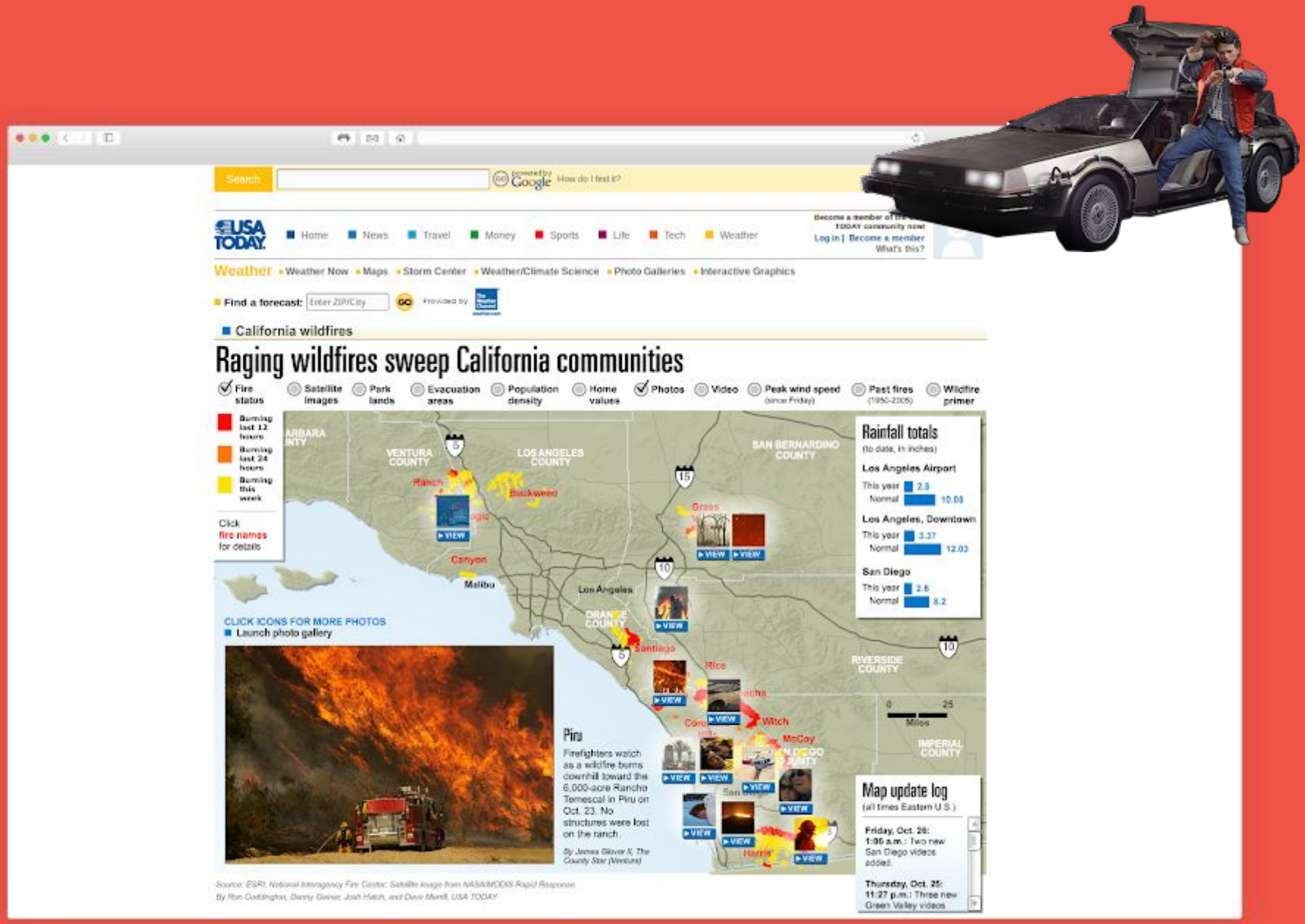


HTML

https://en.wikipedia.org/wiki/HTML



http://designerhacks.com/sketchupgooglemaps/

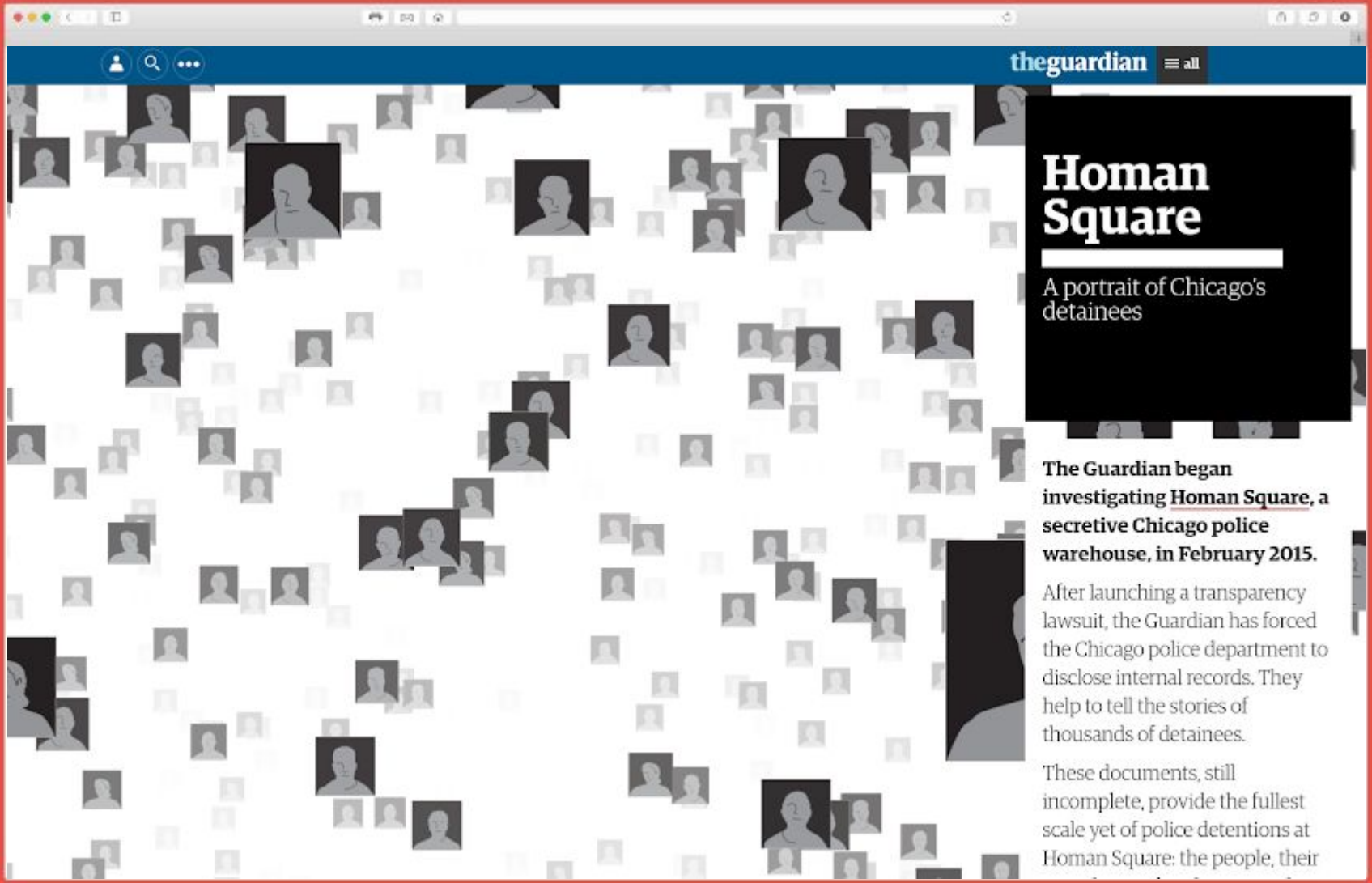


http://usatoday30.usatoday.com/weather/wildfires/2007-10-23-california_fires_N.htm

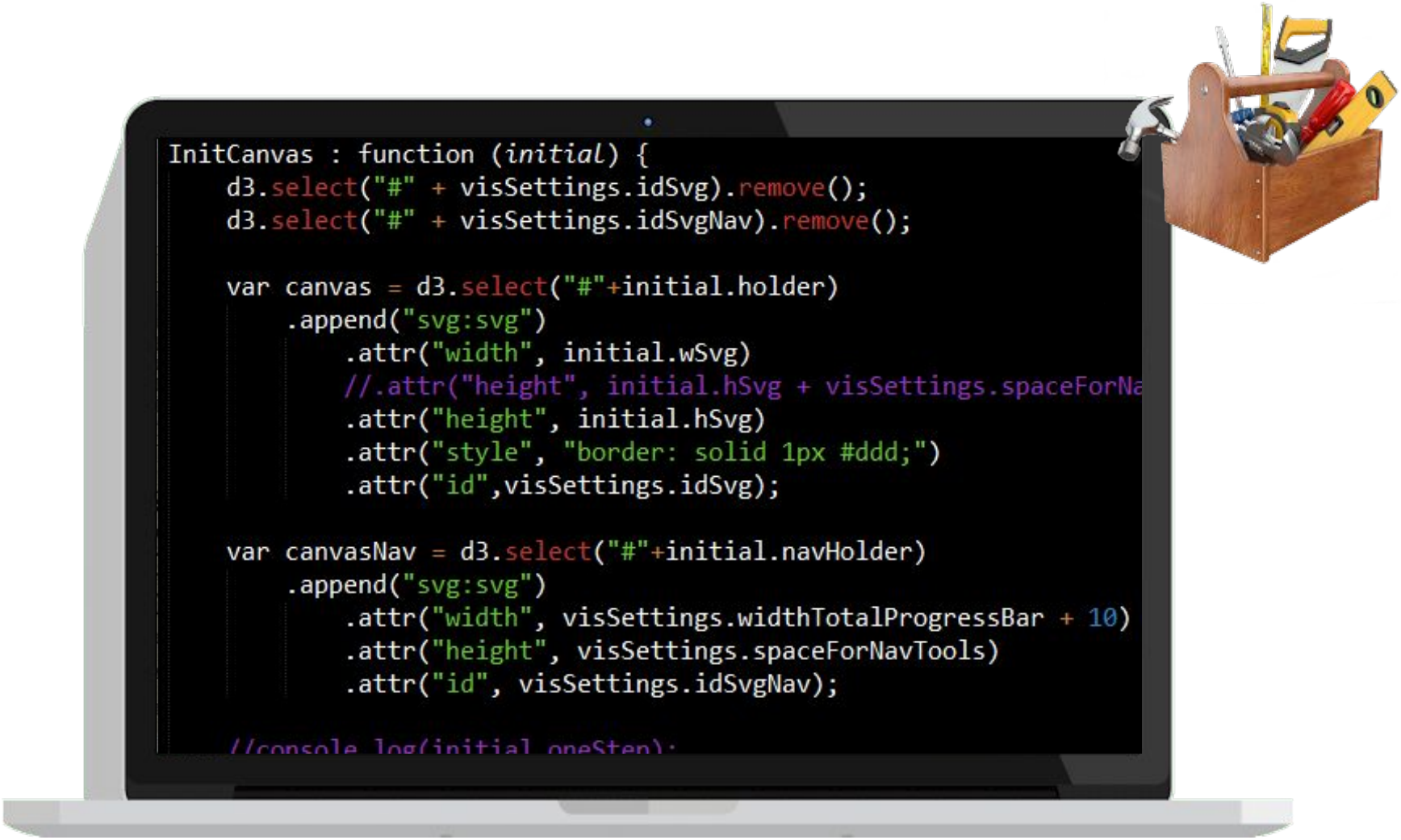
Two:

Miners

min·er, *noun.* **1.** an interactive graphic that gradually reveals a series of concepts by digging through a data set.



https://www.theguardian.com/us-news/ng-interactive/2015/oct/19/homan-square-chicago-police-detainees



JavaScript

https://en.wikipedia.org/wiki/HTML

Jim Vallandingham

Posts Visualizations Experiments About

So You Want to Build A Scroller

Created: 08 April 2015

A critical component of successful interactive visualizations is orienting the users to what they are looking at. In a multi-faceted data story, it can be useful to explicitly move a user through the different views of the data.

Previously, we've looked at steppers as a way to implement this guidance system.

Here, I'd like to look at how the simple act of scrolling can be harnessed to direct a user through a complex visualization combined intuitively with a guiding narration.

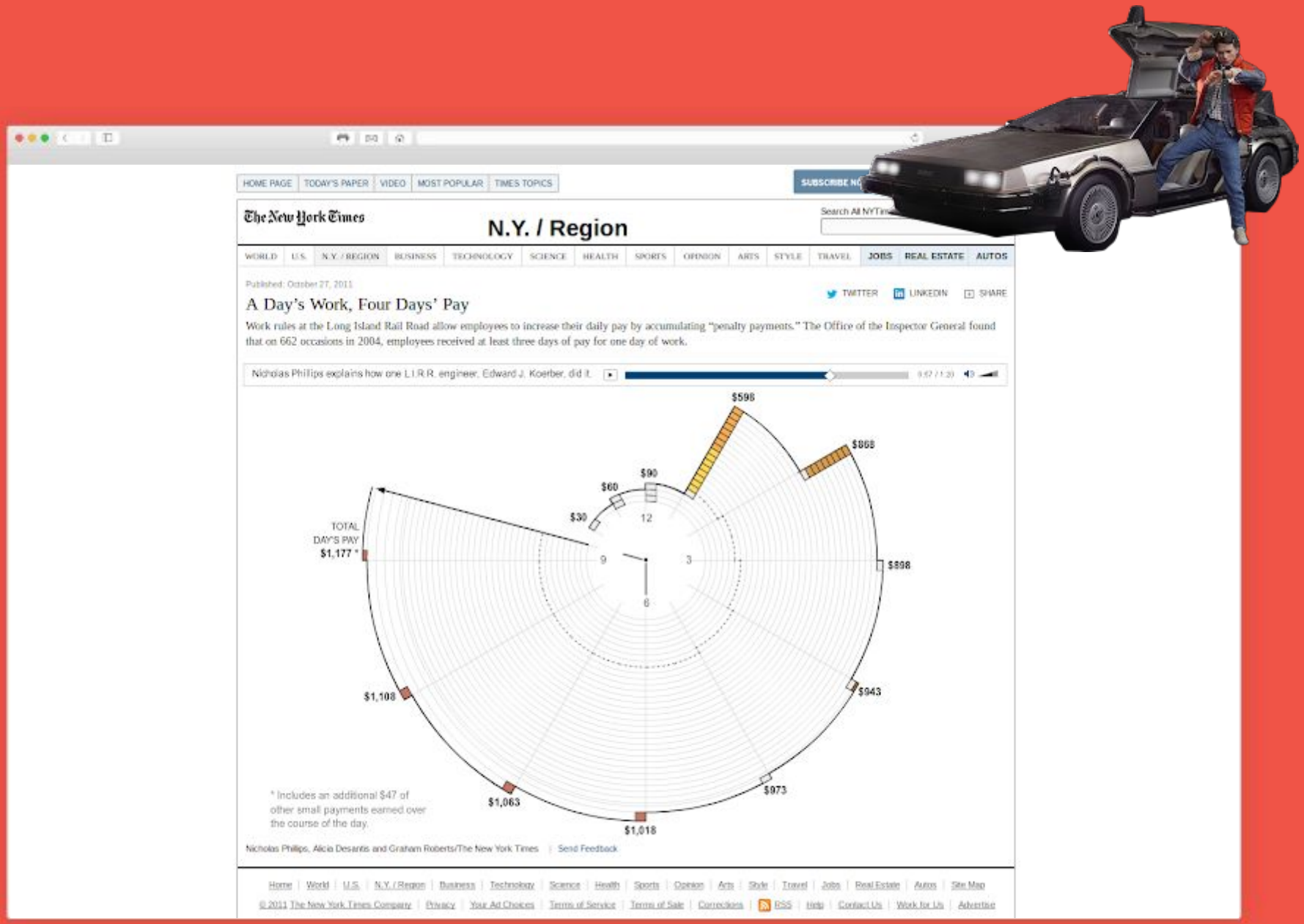
Data Scrolling

There have been many interesting uses of scrolling in data visualization and interactive storytelling (which I will talk a lot more about in my upcoming presentation). In fact, Mike Bostock recently published a wonderful set of best practices on How To Scroll to maximize the use of this tool.

For this tutorial we will look at some of the implementation details for scrolling in data visualizations. I'll focus on a common pattern that moves the user through discrete sections of content, while modifying the visualizations accompanying the text to create a unified data driven story.

This technique has been used in many places including the New York Times with Gregor Aisch's The Clubs That Connect The World Cup:

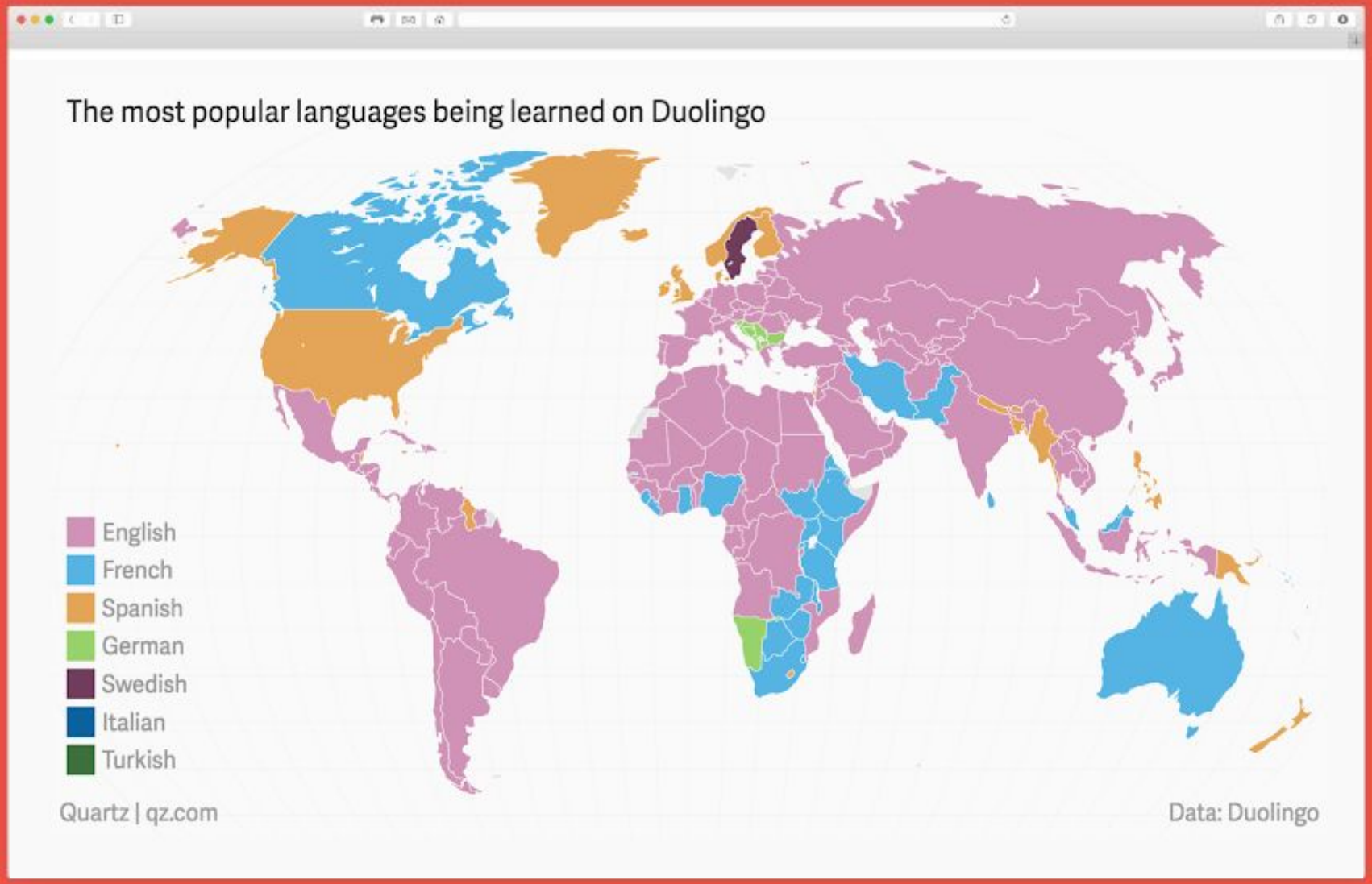
http://vallandingham.me/scroller.html



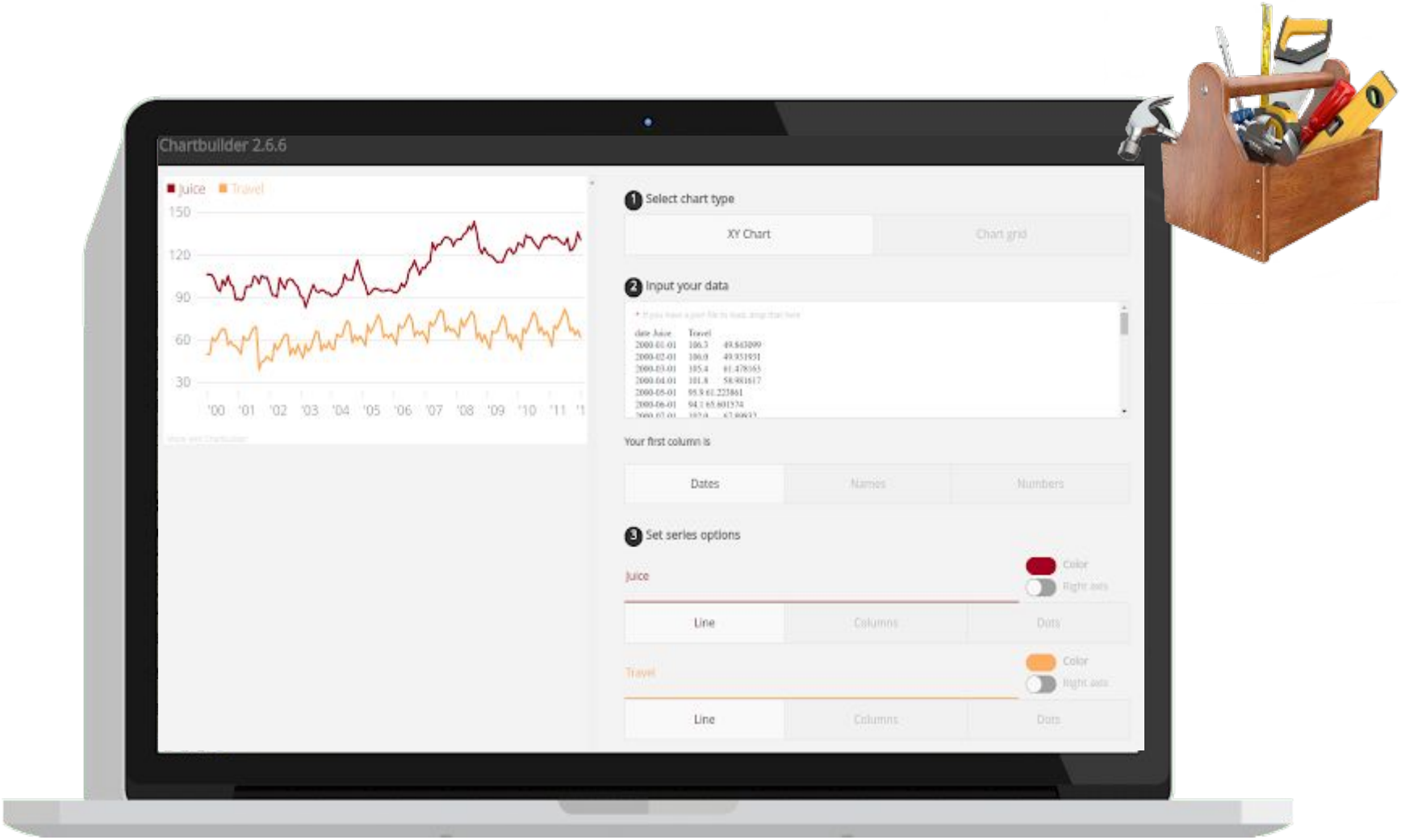
http://www.nytimes.com/interactive/2008/09/19/nyregion/20080920_LIRR_GRAPHIC.html

Three:
Sculptors

***sculp·tor*, *noun*. 1.** a single graphic that stands alone and reveals more to the reader as they careful study its detail.

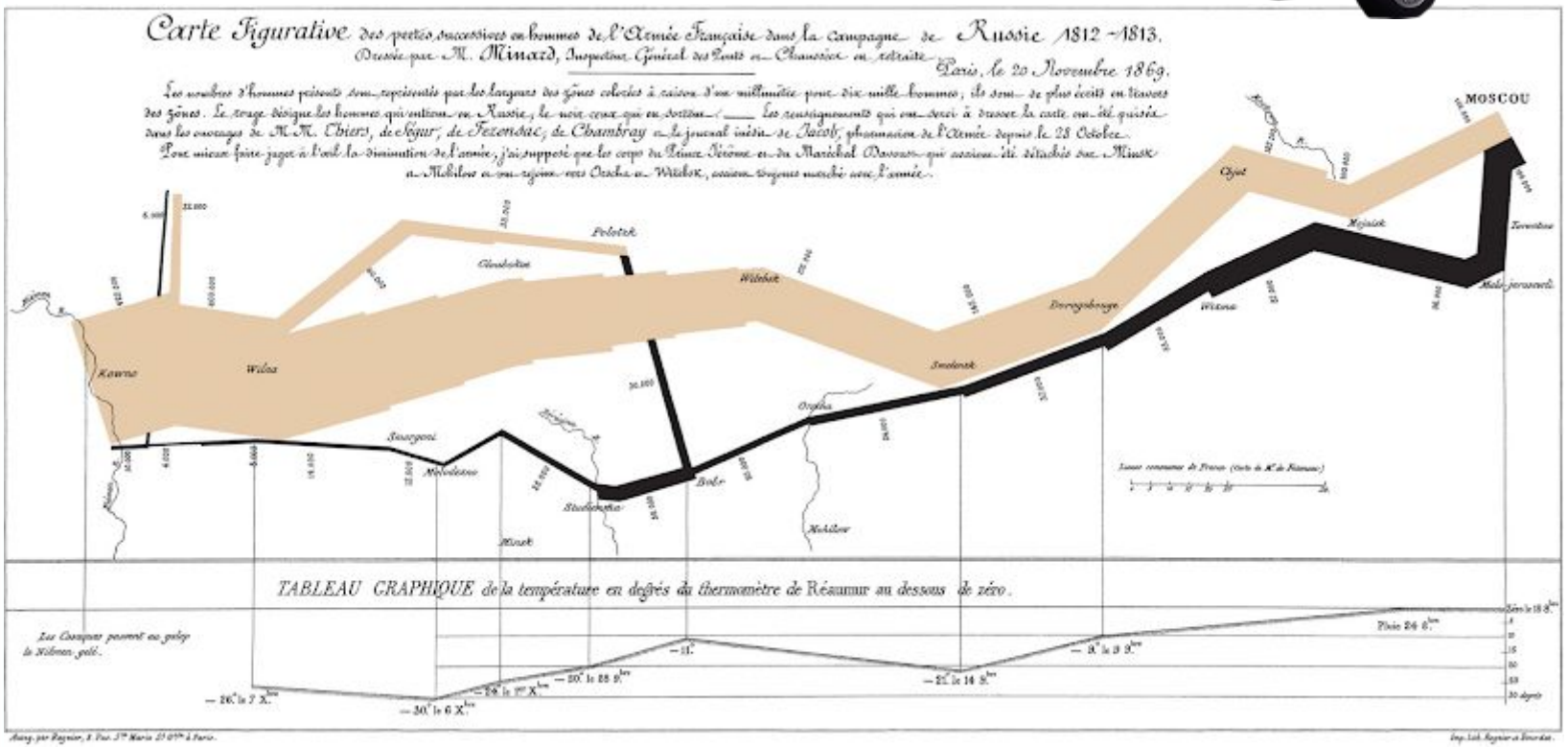


http://qz.com/677833/the-languages-the-world-is-trying-to-learn-according-to-duolingo/



Static chart generators

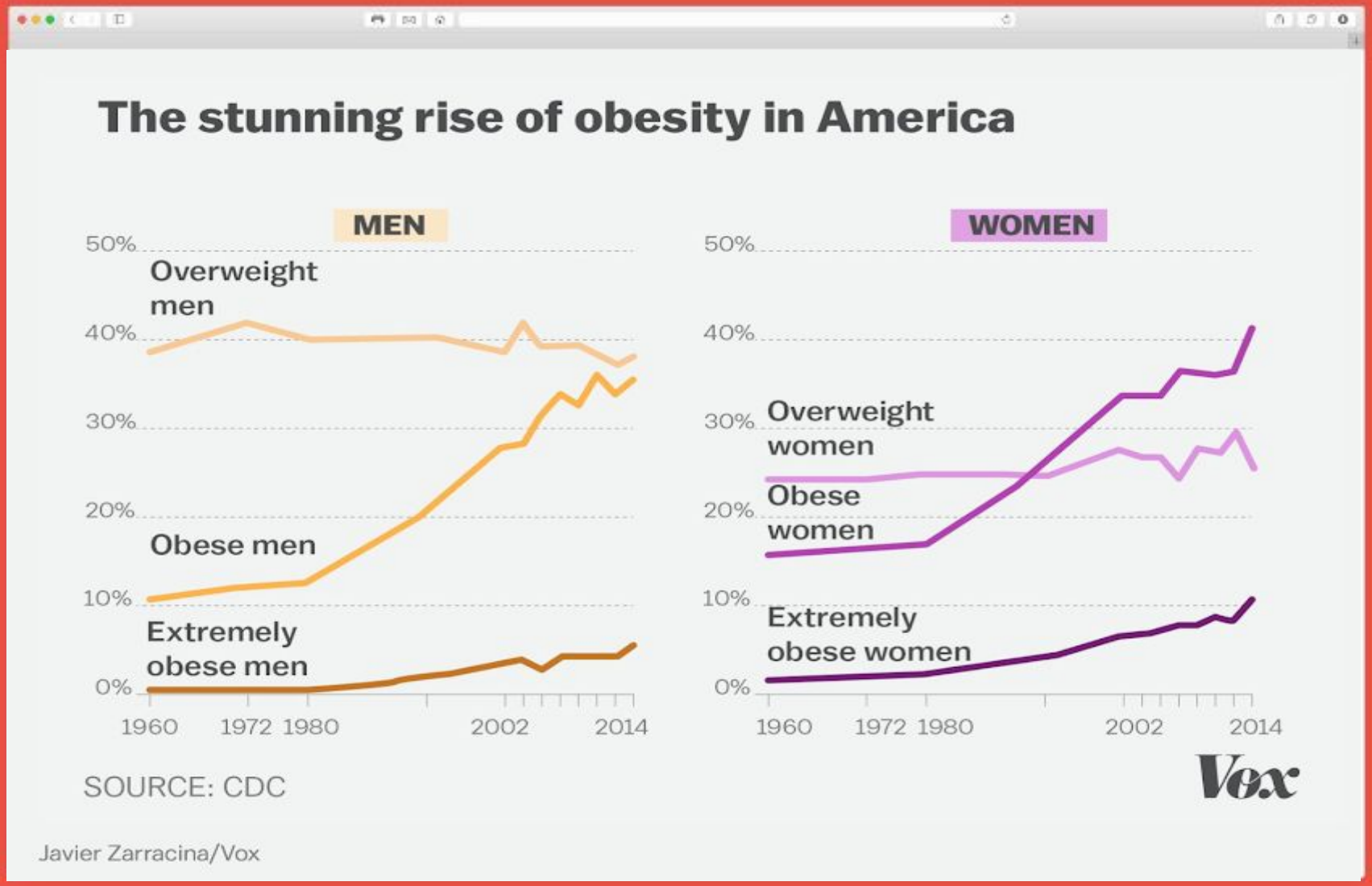
https://quartz.github.io/Chartbuilder/



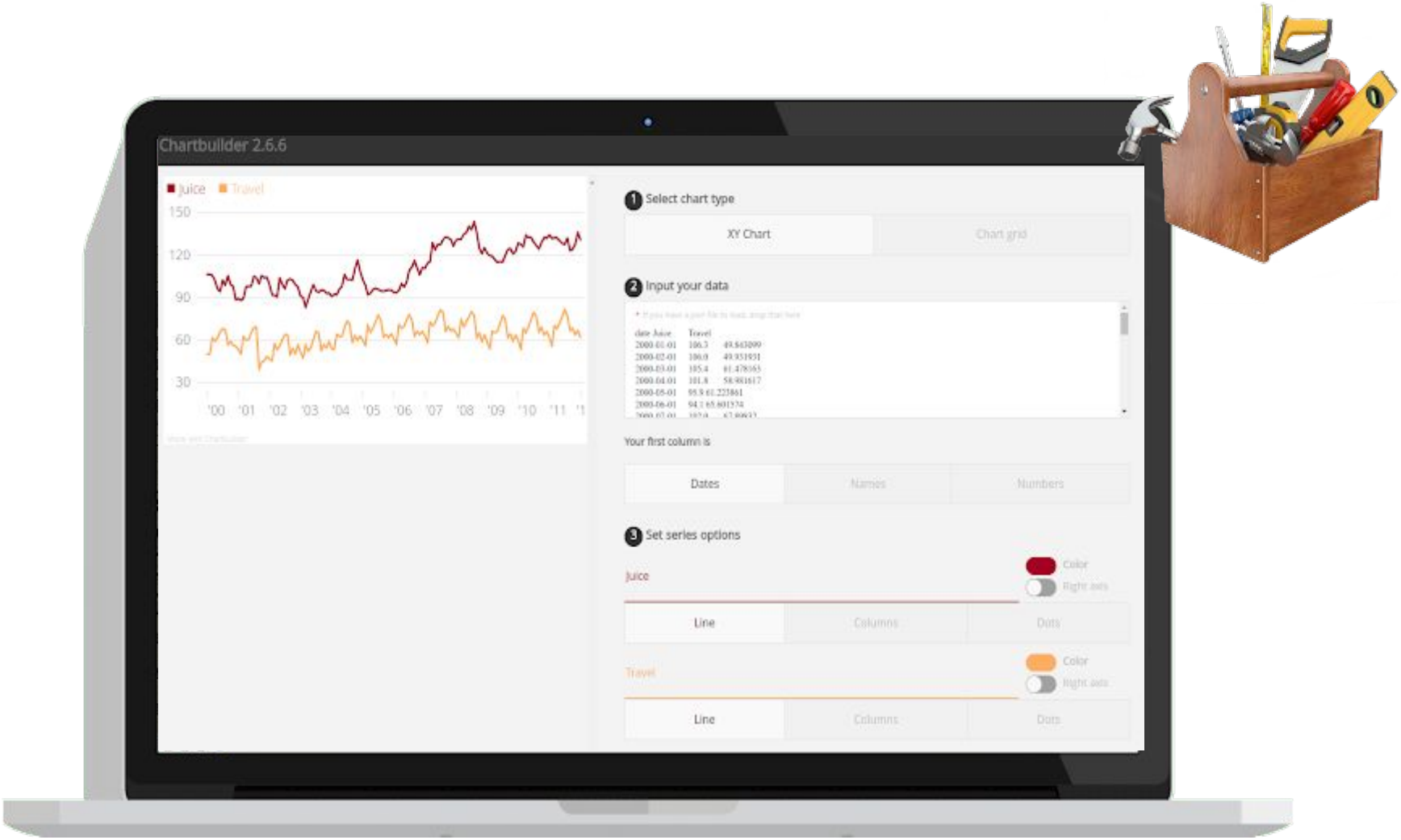
https://en.wikipedia.org/wiki/Charles_Joseph_Minard

Four:
Debaters

de·bat·er, *noun.* **1.** a pointed argument made by interweaving a series of charts within a larger text.

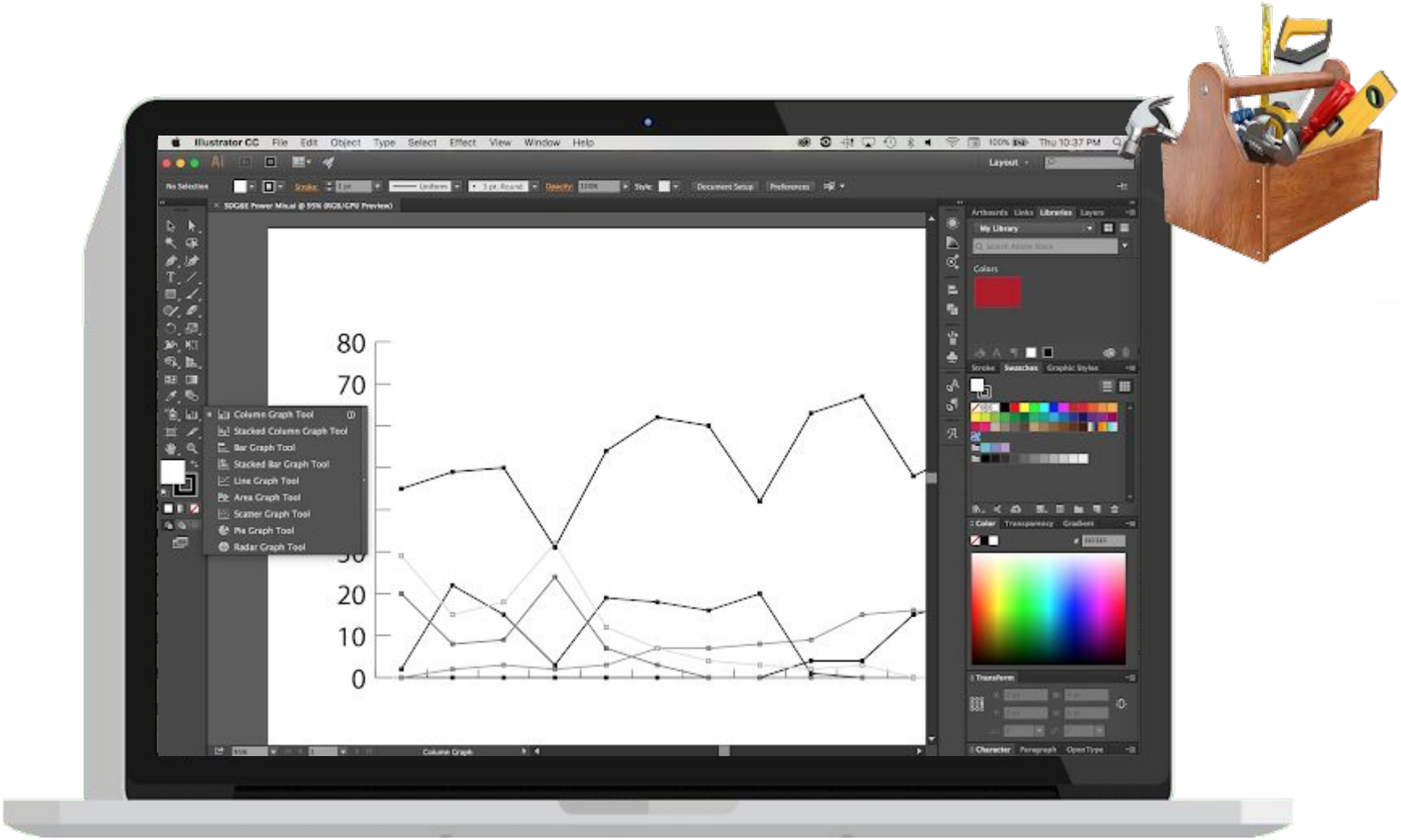


http://www.vox.com/2016/8/31/12368246/charts-explain-obesity



Static chart generators

https://quartz.github.io/Chartbuilder/



Adobe Illustrator

https://quartz.github.io/Chartbuilder/

coursera

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Design and Make Infographics (Project-Centered Course)

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Design and Make Infographics (Project-Centered Course)

About this course: In this project-centered course*, you will create a content-rich Infographic on a topic of your choice using Adobe Illustrator (which you can download for a free, 30-day trial). You might choose to create a visual representation of data from the world of sports, entertainment, politics, or science, to explain a business trend or environmental issue, or even to present a theme or development from your

More

Created by: Michigan State University

Taught by: Karl Gude, Graphics Editor in Residence
School of Journalism

Commitment	6 hours of study, 8 hours of project work
Language	English
How To Pass	Pass all graded assignments to complete the course.
User Ratings	Average User Rating 4.6 See what learners said

https://www.coursera.org/learn/infographic-design

COLORADOAN EDITORIAL BOARD

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WRITE US

The Coloradoan editorial board, which forms the opinions stated in Our Views, usually meets at 9 a.m. Wednesdays. For information, call Josh Awtry, executive editor, at (970) 224-7755.

Letters to the editor are 250 words maximum; public thank-yous are 100 words maximum; Soapbox opinions are a maximum of 550 words. Photos are required for Soapboxes. Soapboxes containing political candidate endorsements will not be accepted, but letters to the editor with endorsements are welcome. Mail, 1300 Riverside Ave., Fort Collins, CO 80524; email, Opinion@coloradoan.com; fax, (970) 224-7899. Phone number and address are required. Entries are edited. Submissions may be published or distributed in print, electronic or other form.

WRITE TO THE EDITOR

TOPIC: ELECTION

Legalizing marijuana not good for us

Ray Martinez
SOAPBOX

Please carefully consider the proposed Colorado constitutional Amendment 64 and Fort Collins marijuana ballot Question 301 that we will vote on in November. Since the amendment doesn't mandate taxing marijuana sales, there is no guarantee there will be any money at all. Banks will not accept the drug money, which

Colorado Educational Association and numerous school boards

ical marijuana card carriers can possess up to 2 ounces of marijuana (120 marijuana cigarettes). According to a story in the Denver Post, "Colorado patients, caregivers and dispensaries all have diverted medical marijuana to illegal use in 23 states," making Colorado "on track to become a primary source of supply for

National Jewish Health have reviewed 30 marijuana-growing operations in Denver, Littleton and Larimer County. They found mold levels at 100 times higher than what is considered safe. In some cases, the level was so high their instruments could not register the levels.

On May 2, survey results re-

David McHaffey
SOAPBOX

Future of United States is education, not football

I am an alumnus of CSU (Colorado A&M, 1954, B.S.) who is adamantly opposed to building a new football stadium.

In this era of transparency, why haven't the donors of this enormous sum of money been identified and what have they been promised in exchange for this donation? It is a completely irresponsible proposal.

I majored in entomology when I attended Colorado A&M, receiving an excellent background for my career in agriculture and entomology and to pursue advanced degrees from other universities. The entomology department no longer exists at CSU. I also played tennis for A&M, but the men's tennis team no longer exists. The proposal to waste millions of dollars (and a large amount of land) to build an on-campus stadium qualifies as a "boondoggle" as described by Dr. David Ridpath (Coloradoan, April 14, 2012). I worked 35 years in 10 states and 10 countries and have never once been asked what the football team

Five:

Tailors

***tai·lor*, *noun*. 1.** a graphic that encourages users to explore and personalize a dataset using searches, filters, quizzes and other interactive tools

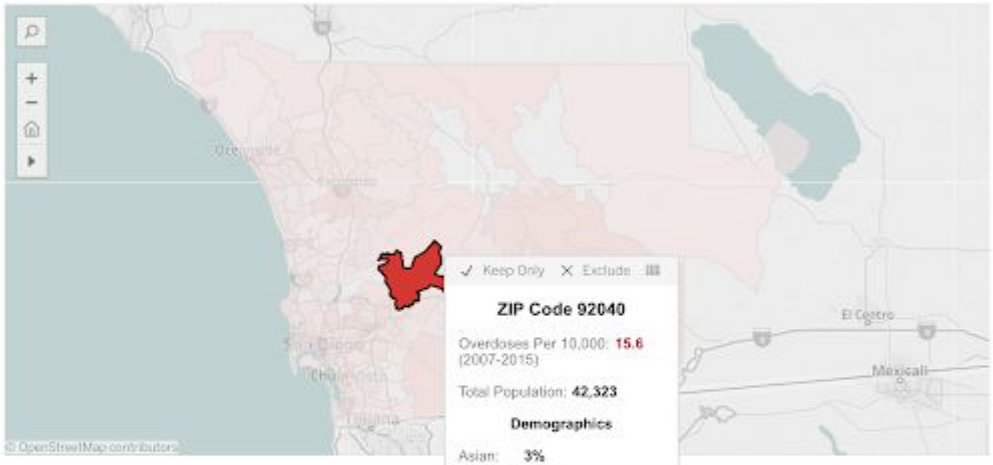
Overdosing in San Diego

Hundreds of people have died of opioid overdoses across San Diego County since 2007. Darker red on the map means a higher number of deaths per for 10,000 residents.

Click on the map or use the search bar to explore how deaths have changed in those ZIP codes.

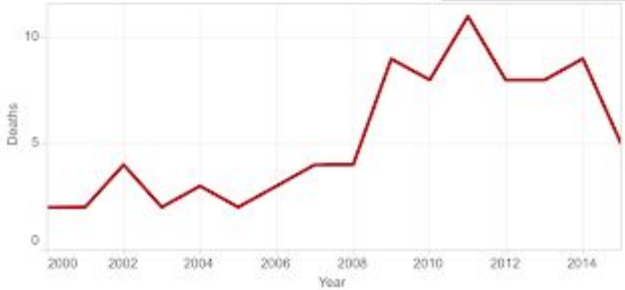
Search for a ZIP code

(All)



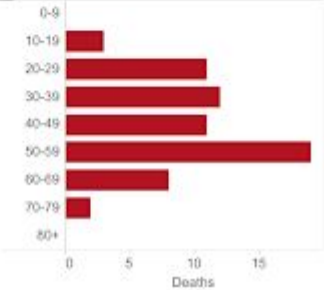
Deaths over time

Overdoses between 2000 and 2015 in this ZIP c



Different age patterns

Ages of overdose victims in this ZIP code.

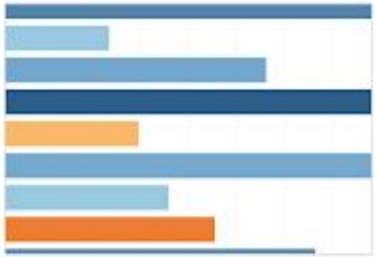


http://inewsource.org/2016/09/12/opioid-san-diego-county/

Tableau

http://www.tableau.com/

Free Training Videos

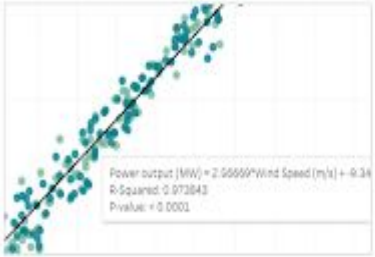


3 VIDEOS

Getting Started

10 VIDEOS

Connecting to Data



20 VIDEOS

Visual Analytics

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THE END

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